

# Machine Learning-Based Real-Time Localization and Automatic Trapping of Multiple Microrobots in Optical Tweezer

Yunxiao Ren<sup>1</sup>, Meysam Keshavarz<sup>1</sup>, Salzitsa Anastasova<sup>1</sup>, Ghazal Hatami<sup>1</sup>, Benny Lo<sup>1</sup>,  
Dandan Zhang<sup>\*1,2,3</sup>

**Abstract**—Optical Tweezer (OT) is an attractive tool for biological studies, which has been used for cell manipulation and tissue engineering. However, the high intensity of laser beam may damage the target biological objects or specimens. To this end, indirect optical manipulation using optical microrobots has become a promising research direction. To enhance the efficiency of indirect manipulation, automatic localization and trapping of multiple microrobots is significant, which paves a way for closed-loop control. For microrobots localization, a modified YOLOv4-tiny neural network model is developed by integrating ConvNext block for feature extraction, while Gaussian modelling is used to optimize the coordinates of the output of the detection head. To avoid expensive manual annotation for model training, self-supervised learning method is employed with Mosaic data augmentation to eliminate the need of collecting a large amount of data. To determine the optimal trapping points for the laser spots, a machine learning-based ellipse detection method is developed based on arc-support groups and k-means algorithm. We conducted experiments to evaluate the effectiveness of the proposed methods. Compared to mainstream object detection algorithms, our proposed microrobot localization method has higher accuracy and enhanced computational efficiency. The proposed ellipse detection method can detect spherical structures of localized microrobots effectively with 96.77% of success rate. The code is open-source and released on: [https://github.com/ClouseYunxiao/MicrorobotsDetection\\_TrappingPoints](https://github.com/ClouseYunxiao/MicrorobotsDetection_TrappingPoints)

## I. INTRODUCTION

Optical tweezer (OT) can be used to manipulate micro-objects with dimension ranging from 50 nm to 10  $\mu$ m [1]. OT can manipulate cells or other biological objects directly. However, the laser beam with high intensity may cause damages to the target objects or influence their viability [2]. Therefore, indirect manipulation of the OT is crucial to avoid damaging the cell by the laser beam. In this paper, we aim to use microrobots as the end-effectors of OT for indirect manipulation of cells or biological objects.

To implement indirect manipulation with high efficiency, trapping multiple microrobots to control the target biological objects in a cooperative manner is desirable [3]. Thanks to time-multiplexing technique, independent traps can be generated from one single laser beam, which is implemented by switching the beam from one trap position to another within a very short time to create parallel traps at different positions [4]. In addition, multiple GPC (Generalized Phase Contrast) based counter propagating-beam traps [5],

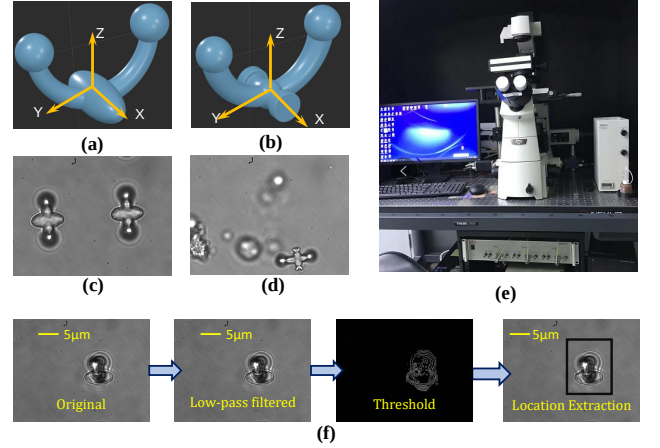


Fig. 1. Overview of the microrobots and the experiments setup. (a) The CAD model of microrobot A; (b) The CAD model of microrobot B; (c) Two microrobots A observed via the optical microscope during for model training; (d) The view of microrobot B observed via optical microscope for model testing; (e) Overview of the experiments setup. (f) Overview of data preprocessing.

holographic OT [6] can be used to reach this target. With these techniques, multiple silica microspheres can be trapped by OT to form a reconfigurable gripper and transport yeast cells in cooperative manner [7]. To enhance the dexterity of indirect manipulation, microtools actuated by OT have been developed to manipulate biological objects indirectly [8].

However, the manual manipulation of multiple microspheres or microrobots is tedious process. To enhance the operation efficiency, automation is worth developing to reduce human operators' workload. To this end, localization and automatic trapping of multiple microrobots in OT is a prerequisite. Many existing works focus on planning and control for automatic cell manipulation [9], including using a decision theoretic based planning approach for real-time path planning [10] and graph search-based algorithms for automated cell transport in microfluidic chamber [7]. However, little attention has been devoted to real-time localization and automatic trapping of microrobots with complex shapes.

The localization of microrobots can be formulated as an object detection problem, where deep learning-based approaches have proven to be powerful and achieved fruitful results. For example, faster R-CNN [11] hypothesize object locations based on region proposal algorithms. However, this method is difficult to be applied to real-time operation scenarios for robotic control where high computation speed

<sup>1</sup>The Hamlyn Centre for Robotic Surgery, Imperial College London, London SW7 2AZ, UK (email: y.ren21@imperial.ac.uk)

<sup>2</sup>Department of Engineering Mathematics, University of Bristol, Bristol BS8 1TH, UK. (email: ye21623@bristol.ac.uk)

<sup>3</sup>Bristol Robotics Lab, Bristol BS34 8QZ, UK.

is required, due to its complex pipelines. YOLO [12], another popular object detection method, has the advantage of high accuracy while the computation speed is fast enough to enable its deployment for real-time applications. However, these methods have rarely been applied to robot-assisted optical manipulation. Moreover, previous methods require a large amount of annotated data for model training, which is not practical for applications in optical microrobotics systems where data is limited. Therefore, we aim to use a self-supervised learning method for model training to avoid intensive data labeling, while the data augmentation technique is employed to enhance data efficiency.

To enable automatic trapping and transporting of microrobots stably, it's significant to identify the optimal trapping positions of each microrobot and control the optical traps to trap the microrobots. Based on numerical calculation and experimental studies, spherical and symmetrical shapes are easier to be trapped by OT [13]. Therefore, most of the optical microrobots designed for indirect manipulation in OT have spherical handles, which can be trapped by multiple laser spots to provide dexterous motions. To this end, it's important to estimate the position of the central points of the spherical or elliptical structures of the microrobots stably. Algorithms have been developed to trap micro-spheres automatically in OT system [14]. However, these approaches are difficult to be applied to tracking microrobots, especially those with complex shapes. More robust and versatile approaches should be developed to reach this target.

In summary, this paper aims to investigate deep learning-based method for real-time localization of multiple microrobots simultaneously. The **main contributions** of this paper are listed as follows.

- 1) A deep neural network model is constructed based on a modified YOLOv4-tiny architecture for multiple microrobots localization, where a ConvNext-based block is used as the backbone for feature extraction and Gaussian modeling is used to optimize the output of the detection head. Self-supervised learning with Mosaic data augmentation is used for model training.

- 2) A machine learning-based ellipse detection technique is proposed to identify the optimal trapping points for each localized microrobot, through which the positions of the laser spots can be determined to manipulate the microrobots stably.

## II. METHODOLOGY

### A. Database Construction

- 1) *Microrobot Design and Data Collection:* Two different types of microrobots (Microrobot A and Microrobot B) are designed and fabricated for evaluation of the proposed method in this paper. Both Microrobot A and Microrobot B are printed with 18 different out-of-plane poses which is from  $5^\circ$  to  $90^\circ$ , and each different pose has  $5^\circ$  interval. And the CAD models of the microrobots are shown in Fig.1 (a) and Fig.1 (b) respectively.

To achieve the target of localizing multiple microrobots automatically, images of Microrobot A in OT are collected

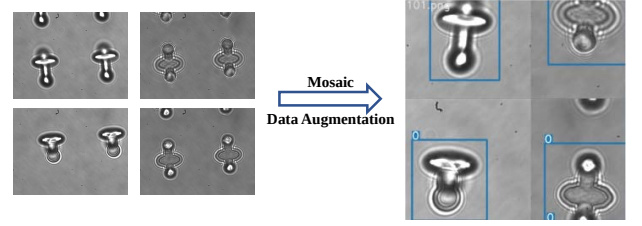


Fig. 2. Overview of the Mosaic data augmentation of train set 1. Four images of microrobot A are mirror mapped, and then cropped and stitched at random positions to form a new image.

for training and validating the deep learning-based model (see Fig.1 (c)). To demonstrate the generalizability of the proposed method, Microrobot B is used for experiments in the testing stage (see Fig.1 (d)), which has slightly different structure compared to Microrobot A. All microscopic images for both the training and testing database were collected using a CCD camera. The training and validating dataset includes 12079 images of microrobot A, while the testing dataset includes 7301 images of microrobot B. 80% of the images in the training database were used for model training, which we call ‘train set 1’ in this paper, while the remaining 20% of images were used for evaluation. During experiments, multiple microrobots can be trapped, and the motions and poses of each microrobot can be controlled by changing the positions of the laser spots.

- 2) *Data Preprocessing:* Most of the existing object detection algorithms are developed based on supervised learning. The data labeling process is tedious and time-consuming. To this end, we develop a method to obtain the 2D position of microrobot A and generate labels automatically for model training, which leads to self-supervised learning.

During data preprocessing, the images of microrobots A were gray-scaled and filtered by a low-pass filter for denoising. Following that, edge detection based on Canny algorithm [15] was applied to get the 2D location information of microrobots. Gaussian noises were added to the images to increase background complexity, which can enhance the robustness of the model for microrobot localization. The data preprocessing process is illustrated in Fig. 1(f).

- 3) *Data Augmentation:* To enhance the diversity of the database and reduce the need of intensive data collection, Mosaic data augmentation method is used in this paper [16]. Four images of the microrobot A in train set 1 were randomly selected first and they were mirror mapped, then the four images were cropped and stitched at random positions to generate a new image, as shown in Fig. 2.

With 2D location information extracted through data preprocessing, new location information of microrobot A is responded in the new image. Eventually, we generated a new database containing 120 images with microrobot A obtained by Mosaic data augmentation as another training set, which named as ‘train set 2’ in this paper. Comparing to train set 1, train set 2 has a smaller amount of data and a more complex background environment, which is more conducive to the

extraction of microrobots' features and differentiating the background.

### B. Microrobots Detection

1) *ConvNext-Based Block*: ResNet, DenseNet and Transformer have been widely used as network feature extractors for downstream tasks such as classification, regression [17], [18]. More recently, inspired by Swin Transformer, ConvNext has been proposed based on the modification of ResNet 50 [17]. It is a pure convolutional model and compete favorably with most of the state-of-the-art Transformers-based structures in terms of accuracy [19].

A ConvNext-based block is proposed in this paper, the structure of which is shown in Fig.3 (b). To prevent loss of edge features and to control the size of the output feature map, five-pixels padding operation is adopted in the ConvNext-based block. Subsequently, a separable depth-wise convolutional layer with stride set as 7 and a convolutional layer with  $1 \times 1$  filter are adopted. The stride of last convolutional layer is set as 7 as recent studies have proved that large size of kernel could enhance the receptive ability of the network [20], [21]. To increase the depth and extraction capability of the block, the ConvNext-based block also uses a residual architecture containing a  $7 \times 7$  deep separable convolution and three normal convolutional layers of size  $1 \times 1$ .

2) *Network Architecture*: Current one-stage target detection algorithms are structured with three components: the Backbone, the BottleNeck and the Detection Head. The overview of the architecture of the proposed network for microrobot localization is shown in Fig.3 (a), which is inspired by YOLOv4-tiny [16]. The backbone network consists of an input layer, a convolutional layer and three stages ConvNext-based block. The input size of the network is set as  $416 \times 416 \times 3$ . A three stages ConvNext-based block is adopted to extract the key features of the microrobot.

CSPDarknet53-tiny is the backbone network of YOLOv4-tiny [16], which is one of the most effective networks for real-time object detection with optimal speed and accuracy.

Compared to CSPDarknet53-tiny, the backbone network of our proposed method has less computational complexity and training parameters, which further enhances the efficiency of the network. The bottleneck of the proposed network adopts the Feature Pyramid Network (FPN), whose structure is the same as that in YOLOv4-tiny [16]. A convolutional layer is connected after the backbone network. The output of the convolutional concat the first ConvNext-based block after a convolutional layer and up-sampling operation. As for the Detection Head, the number of filters in last convolutional layer is set as 18 in this research. Separable convolution operation is adopted to reduce training parameter and ensure operation efficiency. The training parameter of the proposed model are 30.8 million, which is 7.4 million less than YOLOv4-tiny.

3) *Gaussian Modelling and Loss Function*: The prediction of BBox in YOLO-series contains 4 elements:  $x$ ,  $y$ ,  $w$  and  $h$ , where  $x$ ,  $y$  represent the centroid coordinate of the BBox, and  $w$ ,  $h$  represent the width and height of the prediction of BBox. To enhance the reliability of the prediction of Bounding Box (BBox), Gaussian modelling approach is used in this paper [22]. By incorporating Gaussian modeling, eight elements are predicted instead, including  $\mu_{tx}, \Sigma_{tx}, \mu_{ty}, \Sigma_{ty}, \mu_{tw}, \Sigma_{tw}, \mu_{th}, \Sigma_{th}$ , where  $\mu$  represents the coordinates of BBox, and  $\Sigma$  represents the uncertainty of each coordinates.

The original loss functions in YOLOv4-tiny contain three loss functions of the BBox coordinates loss, confidence loss, and predicted category loss. The loss functions of proposed method are shown below:

$$L_x = -\sum_{i=1}^W \sum_{j=1}^H \sum_{k=1}^K \gamma_{ijk} \log (N(x_{ijk}^G | \mu_{tx}(x_{ijk}), \Sigma_{tx}(x_{ijk})) + \varepsilon)) \quad (1)$$

$$L_{CIoU} = 1 - IoU + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \alpha v \quad (2)$$

$$Loss = L_x + L_y + L_w + L_h + L_{CIoU} \quad (3)$$

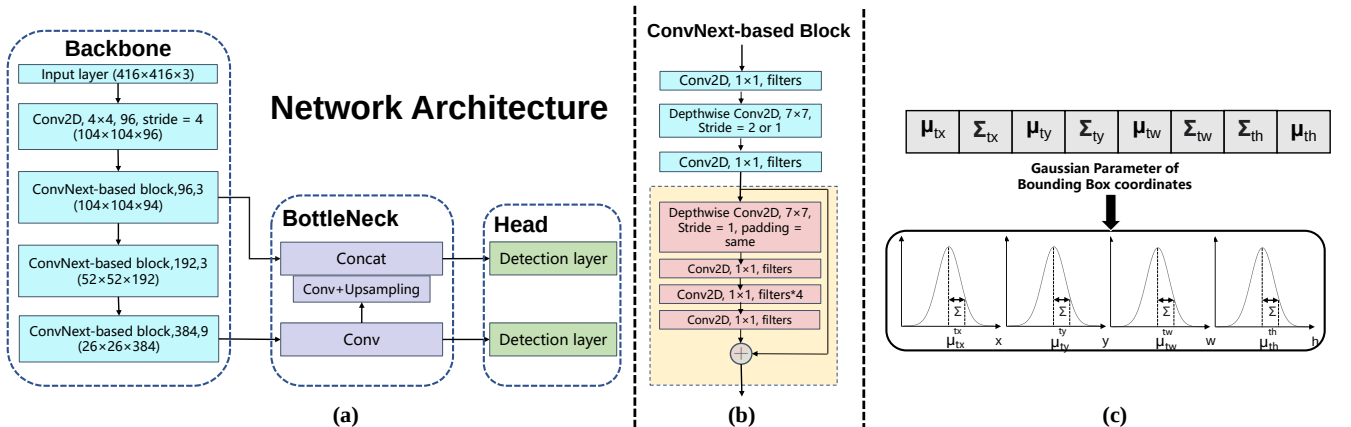


Fig. 3. (a) The overview of the network architecture; (b) The overview of the proposed ConvNext-based block; (c) Overview of Gaussian modelling, where  $\mu$  represents the coordinates of Bounding Box (BBox), and  $\Sigma$  represents the uncertainty of each coordinate.  $x$  and  $y$  represent the coordinate of the centroid of BBox;  $w$  and  $h$  represent the width and height of predicted BBox.



$L_x$  represents the loss of  $t_x$  coordinates, and the others (i.e.,  $L_y, L_w, L_h$ ) are the same as  $L_x$  except for each parameter.  $W$  and  $H$  represents number of grids of each *width* and *height*;  $K$  means the number of filters in last convolutional layer mentioned before [22].  $L_{CIOU}$  is the confidence loss which adopted CIOU loss [23], same as that in YOLOv4-tiny. And  $Loss$  is the total loss of the proposed algorithm during training.

### C. Determination of Optical Trapping Points

After detecting and localizing the microrobots in the image plane, the following stage is to trap the microrobots automatically, which requires the determination of the optimal trapping points to control the positions of the laser spots.

Though deep learning-based methods can be used to handle ellipse detection as well, expensive manual annotation limits its deployment in real-world applications [24]. To achieve higher computation speed and location accuracy, we use traditional computer vision technique for spheres/ellipsoids localization. The ellipse detection method used in this paper includes four steps, i.e. i) Arcs extraction, 2) Arc-Support groups forming, 3) Ellipse generation and 4) Filtering.

1) *Arcs Extraction*: The smooth curve extraction process is performed when the change in the curvature of an ellipse is continuous. The process is performed by using the smooth function known as the Gaussian kernel. Subsequently, a Canny edge operator [15] is used to extract the edge points from the gray-scale image. In this step, an eight-neighbor algorithm proposed by Wang et al. [25] is adopted to get continuous edge contours in this paper.

After extracting the continuous edges, the corners (also known as vertices) must be split into two parts. The first step involves performing a curvature calculation and a Gaussian filter to split the edges. The difference between the standard deviation and the curve's smoothing effect is then taken into account to determine the optimal filter.

The selected closed edges are then fitted and regarded as the final detection. Then the adaptive threshold  $T_{corner}$  (see (5)) and the curvature are used to determine the locations of the corners along each edge. The formulas below show the curvature calculation:

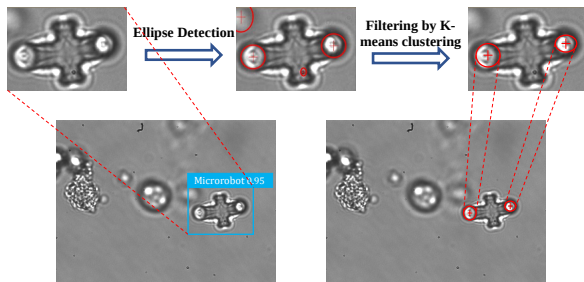


Fig. 4. Illustration of the workflow for microrobot localization and optimal trapping points determination.

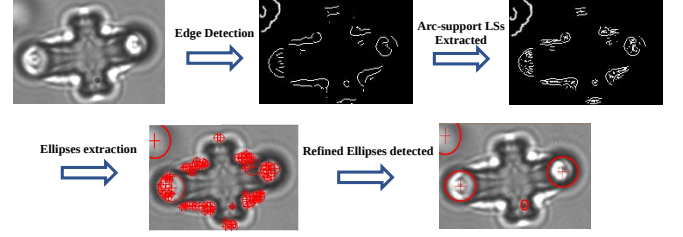


Fig. 5. Overview of the elliptic extraction process.

$$\kappa_i(\theta, \sigma) = \frac{\ddot{y}}{\dot{x}^2} = \frac{\sigma}{\sqrt{2\pi}} \cdot \left( e^{-y_s^2/2\sigma^2} - 1 \right) + \frac{y_s}{\sqrt{2\pi} \cdot \sigma} \left( y_s - \frac{y_s^3}{6\sigma^2} \right) \quad (4)$$

$$T_{corner} = \alpha \cdot \kappa(\theta_{max}, \sigma), \alpha \in [1, 2] \propto \frac{1}{L} \quad (5)$$

$\theta$  represents the angle of negative correlation in the corner (the detailed definition can be found in [25]);  $\dot{x}$  and  $\ddot{y}$  represent the first and the second derivatives of the coordinates in (4).  $\theta_{max}$  is usually within the range of 90 to 120 degrees. However, due to the length of the edge's contours except for the influence of  $\sigma$ , it changes. They are then split into continuous edges and removed for a purer elliptic arc. Fig.5 shows an example of elliptic extraction in the microrobot image.

2) *Arc-Support Groups Forming*: An elliptical curve can contain several arcs-support line segments (LSs) [26]. With arcs extracted and some of the arcs-support LSs linked together, the resulting group should meet the convexity and continuity conditions. The head of an arc-support LS should be closed enough to that of another one. The linked LSs should align in the same direction if the condition is convex. The number of supported points of each next LS within a statistical area near its terminal is then computed to show the maximum number of votes that can be cast for the next LS. This method is performed to create a histogram that shows the optimal number of votes for the next LS.

A group composed of several arcs-support LSs is referred to as an arc-support group. If the number of supports points in the group is  $n$ , then its angle intervals will be derived from the two consecutive arcs-support LSs. The spanning angle of an arc-support group is computed by assuming that its angle interval sequence is  $\{\theta_1^i, \theta_2^i, \dots, \theta_{n-1}^i\}$ , and its salient score is  $\sum_{j=1}^{n-1} \theta_j^i$ . If the group is more salient to an elliptical curve, its spanning angle will be bigger.

3) *Ellipse Generation*: Since an arc-support group might contain multiple arcs, the initial set of ellipses can be generated using two complementary methods. First, the group with the highest salient score is considered the dominant component of the set. The second method is to individually fit the arcs of the group to an ellipse based on the threshold  $T_{ss}$ . This method can be performed globally by searching for the valid pairs of arcs. And all groups should be considered as valid pairs of arcs based on the following criteria: region restriction, adaptive inliers, and polarity constraint.

4) *Filtering*: In addition to spherical structures of the microrobots, other micro-particles in the environment may be detected as ellipses by mistake. Therefore, k-means algorithm is used to cluster the data points and determine the centres of the desired positions for trapping [27].  $k=13$  is used in this paper.

### III. EXPERIMENTS AND RESULTS

#### A. Experiment Setup and Data Collection

A 3D printing system (Nanoscribe GmbH, Germany) was used to fabricate the microrobots based on Two-photon polymerization (2PP) [28]. Photoresist (Nanoscribe, IP-L 780) is selected as the material for microrobot fabrication. After printing the microrobots on the glass slide, the microrobots are placed in deionized water within a spacer as a sample for experiment. The glass slide was then placed on the sample holder of an OT system (E3500, Elliot Scientific, UK). A CCD camera (Basler AG, Germany) was used to observe the microrobots and record the video data during optical manipulation. The whole experiments setup is illustrated in Fig.1(e).

#### B. Microrobot Detection

With stochastic descent optimization with a momentum of 0.9, and the whole training iteration is set as 6000, and a weight decay of 0.0005, and the initial learning rate is set as 0.001.

To demonstrate the effectiveness of the proposed detection method and small data volume training based on data augmentation, comparisons are made with some mainstream object detection algorithms, including Mask-RCNN, Faster-RCNN, YOLOv3, Gaussian YOLOv3, YOLOv4-tiny and YOLOv4. The results for comparisons are shown in Fig.6. Precision is the measure of the detection accuracy of the object detection algorithm; the Recall shows the sensitivities of the detection. Frames Per Second (FPS) is used to evaluate the computation speed of the model, and 'FPR' means the false positive rate which is used to evaluate the false detection of microrobot. The experiments were conducted under different illumination conditions. According to the

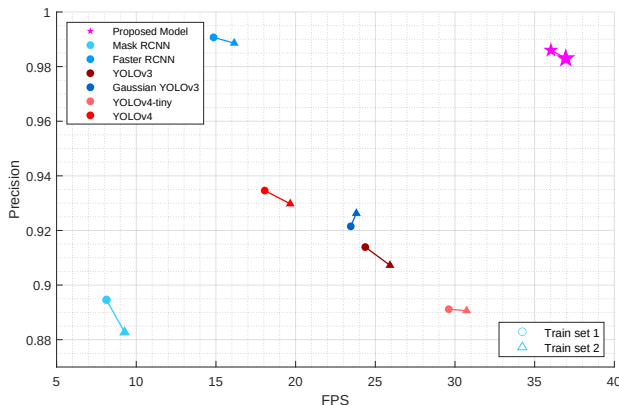


Fig. 6. The experimental results and comparisons with some mainstream object detection algorithms.

TABLE I  
THE COMPARISON BETWEEN PROPOSED MODEL AND BASELINES.

| Models (train set 1)   | Precision | Recall  | FPS    | FPR     |
|------------------------|-----------|---------|--------|---------|
| <b>Proposed Method</b> | 98.6%     | 99.81%  | 36.02  | 1.477%  |
| Mask-RCNN              | 89.46%    | 91.41%  | 8.13   | 17.339% |
| Faster-RCNN            | 99.071%   | 96.77%  | 14.85  | 12.723% |
| YOLOv3                 | 91.391%   | 97.35%  | 24.37  | 6.75%   |
| Gaussian YOLOv3        | 92.15%    | 95.423% | 23.46  | 8.41%   |
| YOLOv4-tiny            | 89.117%   | 94.748% | 29.61  | 7.52%   |
| YOLOv4                 | 93.461%   | 98.552% | 18.06  | 9.614%  |
| Models (train set 2)   | Precision | Recall  | FPS    | FPR     |
| <b>Proposed Method</b> | 97.3%     | 99.931% | 36.95  | 3.114%  |
| Mask-RCNN              | 88.27%    | 89.73%  | 9.27   | 19.227% |
| Faster-RCNN            | 98.86%    | 95.87%  | 16.146 | 13.635% |
| YOLOv3                 | 90.724%   | 96.27%  | 25.92  | 7.126%  |
| Gaussian YOLOv3        | 94.36%    | 95.423% | 23.81  | 10.344% |
| YOLOv4-tiny            | 89.063%   | 96.52%  | 30.732 | 8.91%   |
| YOLOv4                 | 92.977%   | 98.951% | 19.66  | 10.316% |

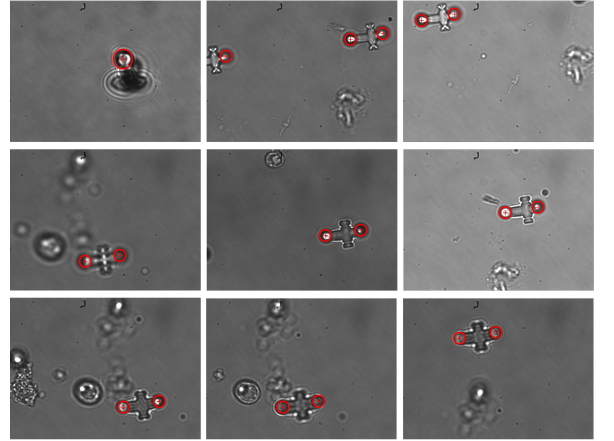


Fig. 7. Overview of some successful examples of trapping points detection in one or multiple microrobots in different environment or with different out-of-plane motion in  $x$ -axis.

experimental results summarized in Table.I and Fig.6, we can conclude that the microrobots can be localized with ease.

All models are trained from 'train set 1' and 'train set 2', respectively, while evaluations of all the models are conducted using test set. Based on the experiment results, Faster-RCNN has better accuracy of the BBox as higher score in precision, the proposed model has better sensitivity of microrobot detection (Recall), less false positive rate. The proposed method has significantly higher computation speed, which indicates that the proposed model is more suitable for real-time detection. By comparing the experimental results using model obtained via 'train set 1' and 'train set 2' respectively, we can conclude that the proposed method has high data efficiency.

#### C. Trapping Points Detection

Fig.4 illustrates the workflow of the proposed method, while Fig. 7 shows successful examples of finding the optimal trapping points of multiple microrobots automatically. To verify the performance of the ellipse detection method, comparisons are made with two baseline models, i.e. Simple

TABLE II

THE COMPARISON BETWEEN PROPOSED ELLIPSE DETECTION METHOD AND BASELINE MODELS.

| Methods                | EED   | EDR    | FPR    |
|------------------------|-------|--------|--------|
| <b>Proposed Method</b> | 27496 | 96.77% | 1.958% |
| Simple Blob Detection  | 21064 | 74.13% | 8.677% |
| Hough Transform        | 23758 | 83.61% | 5.341% |

Blob Detection [29] and Hough Transform [30].

The proposed method has better performance compared to the baseline models according to Table. II, where ‘EED’ represents the effective ellipse detected; ‘EDR’ represents the effective detection rate, and ‘FPR’ represents the false positive rate. For the situations where ellipses can not be detected successfully, the microrobots have out-of-plane pose of 90°. More robust algorithms should be developed in the future to consider microrobots with different out-of-plane poses in the initial state.

#### IV. CONCLUSION AND FUTURE WORK

In this paper, a deep-learning based method is proposed to localize multiple microrobots while ellipsoid detection method is employed to detect the optimal trapping positions of microrobots, which paves a way for automatic indirect manipulation of cells via OT.

For microrobots localization, the proposed model combines the advantages of ConvNext block and YOLO-series algorithms. As for optimal trapping points determination, the ellipse detection method can be used, which is implemented by arcs extraction first, followed by aggregating the arcs and fitting the ellipse, while the k-means algorithm is used to cluster the data points and determine the optimal trapping points in the end. Results indicate that the proposed model can determine the desired trapping points and support the automatic trapping of multiple microrobots in OT. In the future, We envision that the proposed method can be applied to automatic indirect cell manipulation in a microfluidic chip.

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